Pricing and Valuation

Implied forward rate

FRA: $(1+Z_A)^A \cdot (1+IFR_{A,B-A})^{B-A} = (1+Z_B)^B$

Storage cost

$$V_t(T) = \left[S_t - PV_t(I) + PV_t(C)\right] - F_0(T)(1+r)^{-(T-t)}$$

dividends

$$F_0(T) = \left[S_0 - PV_0(I) + PV_0(C)\right](1+r)^T$$

$$c + X(1+r)^{-(T-t)} = S + p$$

$$\begin{tikzcd}
 & \frac{S_1^u}{S_0} \\
S_0 \arrow[ur, "u"] \arrow[dr, "d"'] & \\
 & \frac{S_1^d}{S_0}
\end{tikzcd}$$

$V_0 = hS_0 - c_0 \rightarrow V_1^u = hS_1^u - c_1^u$

$V_0(1+r) = V_1$ → Perfect hedge

$$h = \frac{c_1^u - c_1^d}{S_1^u - S_1^d}$$

$$C_0 = \frac{\pi c_1^u - (1-\pi)c_1^d}{1+r} \quad \text{where} \quad \pi = \frac{1+r-d}{u-d}$$